Here's the complete English translation of the EG2134 Three-Phase Independent Half-Bridge Gate Driver Chip Datasheet (Rev. 1.0) — formatted to match the original structure and technical terminology:

EG2134 Chip User Manual

Three-Phase Independent Half-Bridge Driver Chip

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REV 1.0

EG2134 Chip Datasheet V1.0

Three-Phase Independent Half-Bridge Driver Chip

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Version Revision Record

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Version No.	Date	Description
V1.0	October 18, 2017	Initial draft of EG2134 Datasheet

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1. Features

- High-side floating bootstrap power supply design; withstands up to 300 V
- Integrates three independent half-bridge driver channels
- Compatible with 5 V / 3.3 V logic-level inputs
- Supports switching frequencies up to 500 kHz
- Low-side VCC operating range: 4.5 V – 20 V
- Output drive capability: +1.2 A source / –1.4 A sink
- Built-in dead-time control
- Anti-cross-conduction lockout — prevents simultaneous conduction of high-side and low-side switches
- HIN inputs: active HIGH → control corresponding HO high-side output
- LIN inputs: active HIGH → control corresponding LO low-side output
- Package options: TSSOP20, QFN24

2. General Description

The EG2134 is a cost-effective, high-performance gate driver IC designed for N-channel MOSFETs and IGBTs. It integrates on-chip logic input processing, dead-time generation, shoot-through protection, level shifting, pulse filtering, and high-current output drivers.

The EG2134 operates with high-side supply rails up to 300 V and accepts a wide low-side supply range of 4.5 V to 20 V. Built-in lockout prevents both power switches in a half-bridge from conducting simultaneously. HIN and LIN pins include internal pull-downs — inputs float → both outputs held OFF by default. Output stage delivers +1.2 A peak source / –1.4 A peak sink. Available in TSSOP20 and QFN24 packages.

3. Applications

- 3-Phase Brushless DC (BLDC) Motor Drives
- Uninterruptible Power Supplies (UPS)
- Inverters / Motor Controllers
- High-Power Switching Power Supplies

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4. Pin Assignment

4.1 Pinout Diagrams

Figure 4-1: TSSOP20 Pinout

Table

Pin No.	Name	Pin No.	Name
1	HIN1	11	LO1
2	HIN2	12	VS3
3	HIN3	13	HO3
4	LIN1	14	VB3
5	LIN2	15	VS2
6	LIN3	16	HO2
7	VCC	17	VB2
8	GND	18	VS1

Pin No.	Name	Pin No.	Name
9	LO3	19	HO1
10	LO2	20	VB1

Figure 4-2: QFN24 Pinout

Table

Pin No.	Name	Pin No.	Name
1	LIN1	13	HO3
2	LIN2	14	VB3
3	LIN3	15	VS2
4	VCC	16	HO2
5	NC	17	VB2
6	GND	18	VS1
7	NC	19	HO1
8	NC	20	VB1
9	LO3	21	NC
10	LO2	22	HIN1
11	LO1	23	HIN2
12	VS3	24	HIN3

4.2 Pin Description

Table

Pin Name	Type	Function
HIN1, HIN2, HIN3	Input	High-side logic inputs — Active HIGH 0 = High-side MOSFET OFF 1 = High-side MOSFET ON

Pin Name	Type	Function
LIN1, LIN2, LIN3	Input	Low-side logic inputs — Active HIGH 0 = Low-side MOSFET OFF 1 = Low-side MOSFET ON
VCC	Power	Low-side logic / driver supply — 4.5 V to 20 V
GND	Ground	Low-side ground reference
LO1, LO2, LO3	Output	Low-side gate drive outputs — connect to MOSFET gate
HO1, HO2, HO3	Output	High-side floating gate drive outputs
VB1, VB2, VB3	Power	Floating bootstrap supply positive for each phase
VS1, VS2, VS3	Power	Floating supply return / phase node for each half-bridge
NC	—	No Connection / Do not connect

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5. Functional Block Diagram

Figure 5-1: EG2134 Internal Architecture

- Each phase channel: HIN/LIN → input buffer → noise filter → dead-time & lockout logic → level shifter → output driver
- Low-side path includes fixed delay; high-side path includes high-voltage level shifting
- Bootstrap supply powers the floating high-side driver

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6. Typical Application Schematic

Figure 6-1: Standard Drive Circuit

- VCC powered from 15 V
- Bootstrap network: FR107 fast recovery diode + bootstrap capacitor
- Gate series resistors: R4, R5 (with 1N4148 for fast turn-off)
- VB–VS rail rated for 300 V maximum
- Inputs: HINx / LINx from MCU/PWM controller

- Outputs: HOx / LOx → power MOSFET gates
- Decoupling: C1 = 1 µF ceramic capacitor close to VCC/GND

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7. Electrical Characteristics

7.1 Absolute Maximum Ratings

Stresses beyond these ratings may permanently damage the device. Operation at maximum limits for extended periods may degrade reliability.

Table

Parameter	**Symbol**	**Min**	**Max**	**Unit**
High-Side Floating Supply Voltage (VB1–VB3)	VBS	–0.3	300	V
High-Side Floating Ground (VS1–VS3)	VS	VB–25	VB+0.3	V
High-Side Output (HO1–HO3)	VHO	VS–0.3	VB+0.3	V
Low-Side Output (LO1–LO3)	VLO	–0.3	VCC+0.3	V
Low-Side Supply Voltage	VCC	–0.3	25	V
Logic Input Voltage (HIN)	VIH	–0.3	VCC+0.3	V
Logic Input Voltage (LIN)	VIL	–0.3	VCC+0.3	V
Operating Ambient Temperature	TA	–40	+125	°C
Storage Temperature	TSTG	–55	+150	°C
Soldering Temperature (10 s)	—	—	300	°C

7.2 DC & Switching Characteristics

Conditions: TA = 25 °C, VCC = 12 V, CL = 1 nF (unless otherwise noted)

Table

Parameter	**Symbol**	**Test Condition**	**Min**	**Typ**	**Max**	**Unit**
Operating Supply Voltage	VCC	—	4.5	12	20	V
Quiescent Supply Current	ICC	Inputs floating	—	1	5	µA

Parameter	Symbol	Test Condition	Min	Typ	Max	Unit
Input HIGH Threshold	VIH	—	2.5	—	—	V
Input LOW Threshold	VIL	—	–0.3	0	1.0	V
Input Pull-Down Resistance	RIN	—	—	240	—	kΩ
Output Drive						
Source Current (Peak)	IO+	VOUT = 0 V	+1.2	—	—	A
Sink Current (Peak)	IO–	VOUT = 12 V	–1.4	—	—	A
Low-Side Timing						
Turn-On Delay	tON_L	—	—	300	400	ns
Turn-Off Delay	tOFF_L	—	—	100	200	ns
Rise Time	tR_L	—	—	25	200	ns
Fall Time	tF_L	—	—	20	100	ns
High-Side Timing						
Turn-On Delay	tON_H	—	—	220	400	ns
Turn-Off Delay	tOFF_H	—	—	200	400	ns
Rise Time	tR_H	—	—	25	200	ns
Fall Time	tF_H	—	—	20	100	ns
Dead-Time	DT	CL = 0	50	100	300	ns

7.3 Timing Waveforms

- Figure 7-1: Low-Side (LO) Timing — LIN input → LO output propagation & transition definitions
- Figure 7-2: High-Side (HO) Timing — HIN input → HO output propagation & transition definitions
- Dead time ensures both outputs are OFF for the programmed interval to prevent shoot-through

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8. Design Considerations

8.1 VCC Supply

- Recommended operating voltage: 12 V (range: 4.5 V–20 V)
- Decouple with ≥ 1 µF ceramic capacitor placed directly between VCC and GND pins
- For 3.3 V MCU compatibility: inputs are 5 V/3.3 V tolerant — VCC can remain at 12 V
- Bootstrap diodes must be rated for ≥ 350 V and fast recovery type

8.2 Input & Output Interface

- HIN/LIN: Pulled LOW internally — floating inputs = both outputs OFF (safe state)
- Logic: HIGH = ON; LOW = OFF
- Gate Resistors: 10–100 Ω in series with each gate to suppress oscillation
- Bootstrap Capacitor: 100 nF – 1 µF low-ESR ceramic; rated ≥ 50 V
- Frequency Limit: 500 kHz maximum — ensure dead time is longer than overlap risk

.3 Dead-Time Waveform

Figure 7-3: Dead-Time Timing Diagram

- Dead time (DT) is the interval during which both high-side and low-side outputs are forced OFF to prevent shoot-through
- Input transitions cross the 50% threshold; output switching is delayed such that one output falls below 10% before the other rises above 10%
- Internal logic enforces this blanking interval automatically — no external timing components required

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8. Application Design

8.1 VCC Supply Voltage

Select the appropriate drive voltage for your power device:

- N-Channel High-Side MOSFETs: Recommended VDD operating voltage = 10 V – 15 V
- Low-Side MOSFETs: Recommended VCC operating voltage = 4.5 V – 10 V

8.2 Input Logic Requirements & Output Driver Characteristics

The EG2134 integrates: logic input processing, dead-time control, level shifting, floating bootstrap power, and totem-pole output stages for both upper and lower bridges.

- Logic Thresholds: Input HIGH ≥ 2.5 V; Input LOW ≤ 1.0 V
- Interface: Low input current means MCU I/O can drive HIN/LIN directly — no additional buffering needed
- Output Drive Capability:
    - Peak source current: +1.2 A
    - Peak sink current: –1.4 A
    - High-side floating rail rating: up to 300 V
- Propagation Delays:
    - Low-side turn-on: 300 ns | Turn-off: 100 ns
    - High-side turn-on: 220 ns | Turn-off: 200 ns
- Transition Times (CL = 1 nF):
    - Rise time: 25 ns (both LO & HO)
    - Fall time: 20 ns (both LO & HO)

Input/Output Logic & Truth Table

Figure 8-2: Logic Timing Diagram

Table

HIN	LIN	HO (High-Side)	LO (Low-Side)	Operation
0	0	0	0	Both OFF — safe idle
0	1	0	1	Low-side ON
1	0	1	0	High-side ON
1	1	0	0	Both OFF — cross-conduction lockout active

Function Summary:

- HIN = 1, LIN = 0 → HO = 1 (high-side ON), LO = 0 (low-side OFF)

- HIN = 0, LIN = 1 → HO = 0 (high-side OFF), LO = 1 (low-side ON)
- HIN = LIN = 0 → both outputs OFF
- HIN = LIN = 1 → both outputs forced OFF by internal interlock — shoot-through protection is always active

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9. Package Dimensions

9.2 QFN24 Package Dimensions (Corrected)

Table

SYMBOL	MIN (mm)	NOM (mm)	MAX (mm)
A	0.70	0.75	0.80
A1	—	0.02	0.05
b	0.18	0.25	0.30
c	0.18	0.20	0.25
D	3.90	4.00	4.10
D2	1.90	2.00	2.10
e	—	0.50 BSC	—
Ne	—	2.50 BSC	—
Nd	—	2.50 BSC	—
E	3.90	4.00	4.10
E2	1.90	2.00	2.10
L	0.30	0.40	0.50
K	0.20	—	—
h	0.30	0.35	0.40

Notes:

- Terminal plating: NiPdAu (Nickel Palladium Gold)